Double Machine Learning

The Intuition Behind Orthogonalization

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Causal Inference in Economics

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Outline

Overview

The Causal Question

Why a Simple Comparison Fails

Double ML: Clean First, Estimate Second

Data Example

Visual Evidence

Takeaway

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Overview

Summary Setup

Core idea: Double ML uses machine learning to adjust for many confounders (X), while keeping the causal estimate of $D \to Y$ clean and interpretable.

Running example

Treatment D : job training (1/0)
Outcome Y : monthly wages ($)
Controls X : education, experience, region . . .

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What is the question?

Goal | Causal effect

We want the wage gain **caused by** training — not the gap explained by background.

Target estimand:

$$\tau = \mathbb{E}[Y(1) - Y(0)]$$

$Y(1)$: wage if trained; $Y(0)$: wage if not.

Problem: trainees may differ in X (education, experience, motivation . . .).

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Part 1

Why a simple comparison fails

Swipe for the key idea + visuals.

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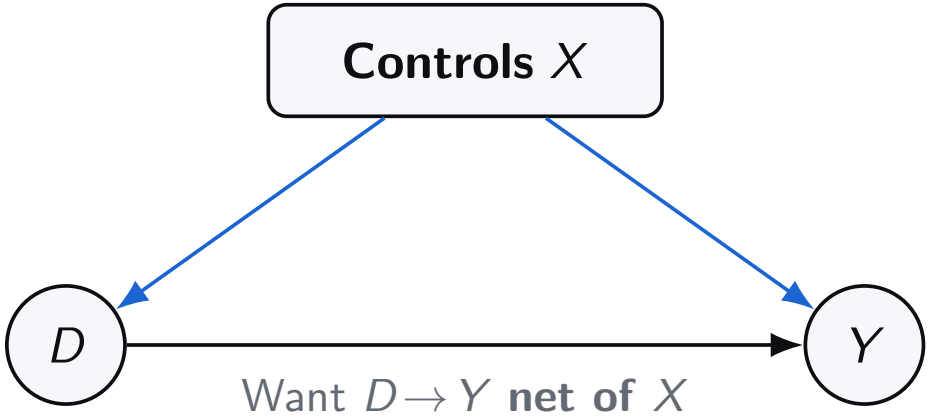
Raw wage gaps mislead

Intuition Bias

- Trainees tend to be more educated.
- Education raises wages regardless.
- So the raw gap **overstates** training.

Naive (biased): $\mathbb{E}[Y|D=1] - \mathbb{E}[Y|D=0]$



Where ML helps (and hurts)

Setup | ML trade-off

Model the world flexibly:

$$Y = \beta D + g(X) + \varepsilon$$

$g(X)$ = complex background effects.

- ML learns $g(X)$ well — great for prediction.
- But regularisation can **distort** β.

Double ML idea: let ML handle background chores, then estimate β on *cleaned* data.

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Part 2

Clean first, then estimate

Swipe for the key idea + visuals.

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Double ML workflow

Pipeline Steps

1. Data

Y wages, D training, X background

↓

2. Two ML models

Learn $\hat{Y}(X)$ and $\hat{D}(X)$

↓

3. Compute residuals

$\tilde{Y} = Y - \hat{Y}(X)$ and $\tilde{D} = D - \hat{D}(X)$

↓

4. Causal regression

$\tilde{Y} = \beta \tilde{D} + \varepsilon$

Key: estimate β from the part of training *not explained by* X.

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The "leftovers" idea

Equations Meaning

Step A — predict with controls:

$$\hat{Y}(X) \approx \mathbb{E}[Y|X], \qquad \hat{D}(X) \approx \mathbb{E}[D|X]$$

Step B — subtract predictions:

$$\tilde{Y} = Y - \hat{Y}(X), \qquad \tilde{D} = D - \hat{D}(X)$$

Causal step: $\tilde{Y} = \beta\,\tilde{D} + \varepsilon$

The slope β *is* the treatment effect.

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Part 3

Data example (wage / labour)

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Illustrative dataset

Data Readable

Y: wage (\$/mo) D: training (0/1) X: educ & exp

Person	D	Educ	Exp	Wage
A	1	16	6	\$3 900
B	1	14	5	\$3 400
C	1	12	4	\$2 950
D	0	16	6	\$3 650
E	0	14	5	\$3 150
F	0	12	4	\$2 700

In practice: X can have hundreds of variables. Double ML scales the same way.

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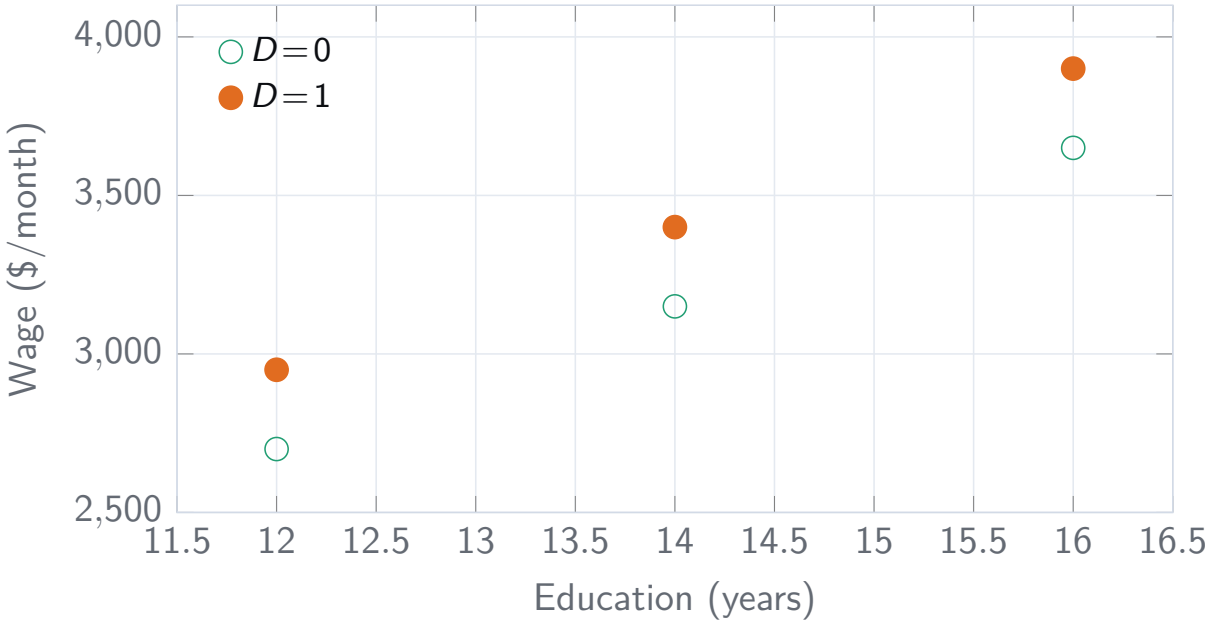
Wages rise with education

Plot | Confounding

- Education predicts wages strongly.
- If trainees are more educated, the raw gap overstates the effect.



Double ML: remove the education / experience part *first*.

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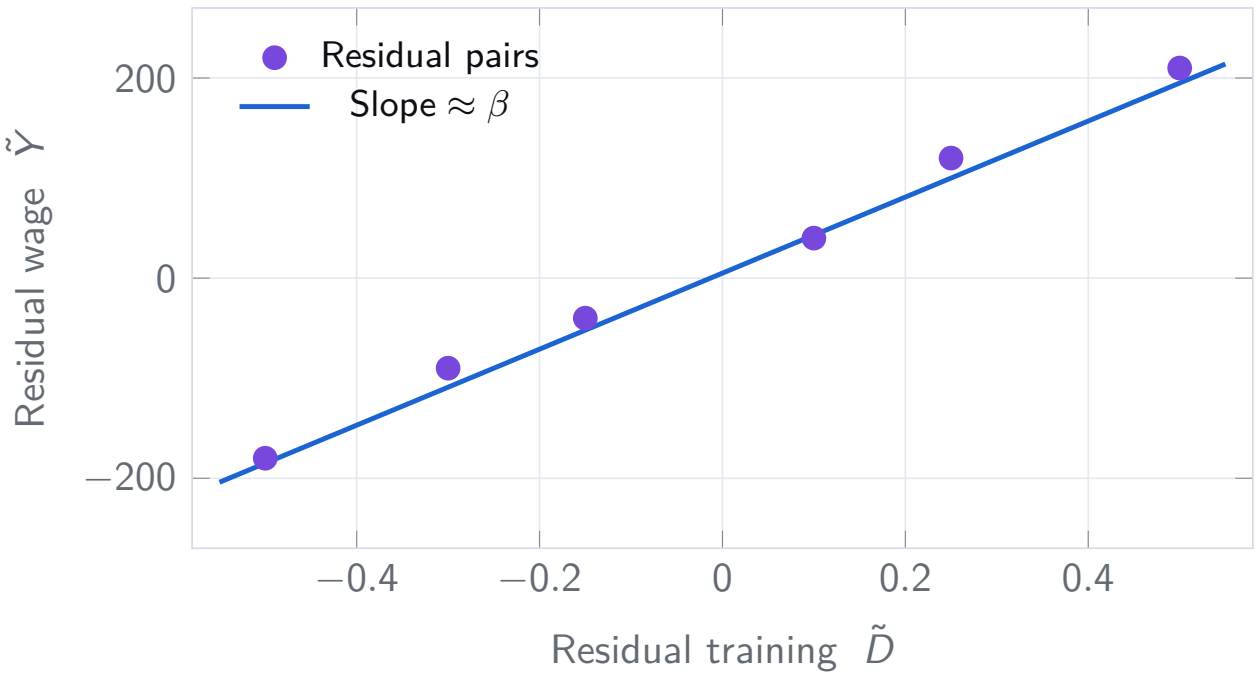
Effect in the left-overs

Residuals | Causal step

After cleaning out X, the slope reveals the causal effect:



$\tilde{Y} = \beta\,\tilde{D} + \varepsilon$ — the slope *is* the causal effect.

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Final take-away

Summary

Double ML uses machine learning to strip out background effects (X), then estimates the causal effect from leftover variation only.

Clean first, then estimate.

Orthogonalisation makes the causal estimate robust to ML prediction errors in the nuisance parts.

End

Thank You

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